

Supporting Information for

Molecular dynamics simulations of liquid silica crystallization

Haiyang Niu ^{a,b}, Pable M. Piaggi ^{c,b}, Michele Invernizzi ^{b,d}, and Michele Parrinello ^{a,b}

^a Department of Chemistry and Applied Biosciences, ETH Zurich c/o USI Campus, Via Giuseppe Buffi 13, 6900 Lugano, Switzerland

^b Facoltà di Informatica, Istituto di Scienze Computationali, and National Center for Computational Design and Discovery of Novel Materials MARVEL, Università della Svizzera italiana (USI), Via Giuseppe Buffi 13, 6900 Lugano, Switzerland

^c Theory and Simulation of Materials (THEOS), Ecole Polytechnique Federale de Lausanne, c/o USI Campus, Via Giuseppe Buffi 13, 6900, Lugano, Switzerland

^d Theoretical Physics, ETH Zurich c/o USI Campus, Via Giuseppe Buffi 13, 6900 Lugano, Switzerland

Local entropy fingerprint

As described in the main text, we employed the recently proposed local entropy fingerprint ($\bar{s}$) [1] to identify atoms with a solid-like environment. We only used the Si atoms to define the fingerprint since we found that it was not necessary to include the O atoms to identify solid-like environments. We calculated the distributions of $\bar{s}$ for the solid (β -cristobalite) and liquid phases at 2300 K in a system with 1536 atoms (see Fig. S1). The distributions of $\bar{s}$ are well separated and therefore the fingerprint can distinguish between solid-like and liquid-like environments remarkably well.

We refer the readers to ref. [1] for more details on how to calculate $\bar{s}$. Here, we only report the parameters that were used. The fingerprint has three parameters, namely, a cutoff r_m for the calculation of the radial distribution function $g(r)$, the width σ of the Gaussian functions used to construct a smooth $g(r)$, and a cutoff r_a to average the fingerprint over the atom's neighbors. We have used, $r_m=0.75$ nm, $r_a=0.45$ nm and $\sigma=0.05$ nm.

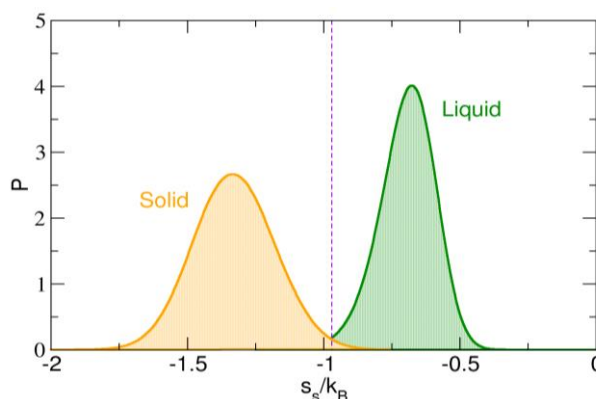


Fig. S1 Distributions of the local entropy fingerprint $\bar{s}$ for silica at 2300 K. Orange and green lines refer to the solid (β -cristobalite) and liquid phases, respectively.

An example PLUMED 2 input for the calculation of the local entropy fingerprint and the subsequent analysis is:

```
# vim:ft=plumed

#####
# Calculation of the fingerprint
#####
# Calculation of the non-averaged fingerprint
# The parameters are r_c=MAXR=0.75 nm and SIGMA=0.05 nm
PAIRENTROPIES ...
  LABEL=s1
  SPECIES=1-1536:24,2-1536:24,3-1536:24,4-1536:24,5-1536:24,6-1536:24,7-1536:24,8-1536:24
  MAXR=0.75
  SIGMA=0.05
  NHIST=201
  LOWMEM
... PAIRENTROPIES

# Averaging of the fingerprint
# r_a=R_0=0.45 nm
LOCAL_AVERAGE SPECIES=s1 SWITCH={RATIONAL R_0=0.425 D_MAX=0.45 NN=50 MM=100}
LOWMEM LABEL=sa

#####
# Analysis
#####
# Histogram of s1
hh1: HISTOGRAM DATA=s1 GRID_MIN=-10.0 GRID_MAX=0.0 GRID_BIN=1000 BANDWIDTH=0.05
STRIDE=1000
DUMPGRID GRID=hh1 FILE=histo1 STRIDE=1000
# Histogram of sa
hh2: HISTOGRAM DATA=sa GRID_MIN=-10.0 GRID_MAX=0.0 GRID_BIN=1000 BANDWIDTH=0.05
STRIDE=1000
DUMPGRID GRID=hh2 FILE=histo2 STRIDE=1000
# Dump of the per-atom fingerprint for visualization
DUMPMULTICOLVAR FILE=MULTICOLVAR-AVG.xyz STRIDE=1000 DATA=sa
# Print all calculated quantities
PRINT STRIDE=1000 ARG=* FILE=COLVAR
```

Interatomic potential

We have also performed the simulations with another interatomic potential[2]. The results showed that the method works also in this case. However, the predicted melting temperature in this case is much too high (over 4,000 K, in agreement with Ref.[2]) as compared with the experimental value (~2,000 K).

References

- [1] P. M. Piaggi and M. Parrinello (2017) Entropy based fingerprint for local crystalline order, *The Journal of Chemical Physics*, 147, 114112.
- [2] S. Munetoh, T. Motooka, K. Moriguchi, and A. Shintani (2007). Interatomic potential for Si–O systems using Tersoff parameterization. *Computational Materials Science*, 39(2), 334-339.